



ROOT CAUSE ANALYSIS

VIRTUAL CONTROL
VMWARE CLOUD FOUNDATION SOLUTIONS

VCF Operations Full Recovery RCA

End-to-end recovery report for VCF Operations after a Windows Update host crash, covering deployment, certificates, fleet, cloud accounts, NSX, and log integration.

RCA

RECOVERY

VCF OPERATIONS

FULL RESTORATION

VCF 9.0

VMware Cloud Foundation

VCF Operations Node — Full Recovery from Windows Update Crash

Prepared by: Virtual Control LLC **Date:** April 8–10, 2026 **Document Version:** 2.1 **Classification:** Internal — Lab Environment **Product:** VMware Cloud Foundation Operations (Aria Operations) **Duration:** Approximately 10 hours across three sessions

PowerShell users: Every bash block in this document that targets a REST API or other client-runnable command has a paste-ready PowerShell sibling block underneath it. For the full substitution table — `curl` → `Invoke-RestMethod`, `base64` auth headers, `here-docs` → `here-strings`, `jq` / `python3 -m json.tool` → `ConvertFrom-Json` / `ConvertTo-Json -Depth N`, etc. — see [POWERSHELL-TRANSLATION-REFERENCE.md](#). Blocks targeting the appliance shell only (`psql`, `service-control`, `vmmon-cli`, `systemctl`, `nodetool`, `ops-cli.py`, edits in `/etc/vmware-vcops/`, `/storage/db/`, `/storage/log/`, NSX admin CLI, ESXi shell) are flagged "Run on the appliance only" with no PS sibling — they cannot run from a Windows workstation.

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1. Executive Summary

Where commands run in this RCA. Recovery touched:

- **VCF Operations appliance shell (SSH to vcf-ops root)** — most service-level work: `service-control`, GemFire configuration under `/usr/lib/vmware-vcops/`, PostgreSQL `psql` to internal DB, `service-watchdog` operations.
- **vCenter UI (browser)** — VCFA OVA redeploy wizard at `https://<vcenter-fqdn>/ui/`.
- **Fleet Manager UI / API** — for re-registration of VCF Operations adapters.
- **Windows workstation (PowerShell)** — REST API verification, `Test-NetConnection` reachability checks.

Replace any credential placeholder like <your-password> with values from your vault.

On the evening of April 8, 2026, a Windows Update on the management workstation abruptly brought down the entire VCF 9.0.1 lab environment. The VCF Operations node required a full redeployment. Over two troubleshooting sessions spanning approximately 8 hours, the node was deployed three times, had its network configuration fixed, hostname changed, all SSL certificates regenerated, GemFire and PostgreSQL configurations updated, and was connected to Fleet Management.

Root Cause: A Windows Update crashed the management workstation, bringing down the nested VCF environment. The VCF Operations OVA was redeployed three times with incorrect settings. On the final deployment, the appliance picked up IP 192.168.1.67 via DHCP instead of the correct static IP 192.168.1.77. It also used the default hostname VMwareCloudFoundationOperationsClusterNode instead of vcf-ops.lab.local. At first boot, the appliance embedded the wrong IP into SSL certificates, Java KeyStores, GemFire config, PostgreSQL connections, Apache HTTPD, and the CASA cluster membership — over ten different locations.

Resolution: The appliance was fixed in place without redeployment. All certificates were regenerated with the correct IP and FQDN. All configuration files and database entries were updated. A secondary IP alias (192.168.1.67) was added as a permanent workaround for CASA's internal admin slice resolution. A DNS record was created for the alias. Fleet Management was connected to fleet.lab.local. A VCF Cloud Account was added with vCenter, vSAN, and NSX adapters collecting data. NSX Manager was recovered from a cold start. VCF Operations for Logs was integrated with vCenter logs flowing via syslog TLS/1514.

2. Environment Reference

2.1 Infrastructure Details

Parameter	Value
Physical Host	Dell Precision 7920
Hypervisor	VMware Workstation 17.x (Nested)
VCF Version	9.0.1.0
Lab Domain	lab.local
Appliance OS	Photon OS 5.0

2.2 VCF Operations Target Configuration

Parameter	Value
Correct FQDN	vcf-ops.lab.local
Correct IP Address	192.168.1.77

Parameter	Value
Wrong IP (from DHCP at deploy)	192.168.1.67
Wrong Hostname (default)	VMwareCloudFoundationOperationsClusterNode
Gateway	192.168.1.1
Subnet	/24 (255.255.255.0)
DNS Server	192.168.1.230
IP Assignment	Static (after fix)
SSL CA CN	VCFOps-cluster-ca_34a44cad-2f3e-4b5e-8c20-a5fae2b908ec
Slice UUID	615ef35e-a1e8-46f0-a1ee-77e9623c9e0a
Root Password	
Keystore Password	KaqYLZxZ48HdioWMG0uUFOJTeJKu4n
Truststore Password	nT3DIApTGJX/Mcm9n7DIH2K2wT5ftd

2.3 Upstream Dependencies

Component	IP Address	Role
vCenter Server 9.0	192.168.1.69	Infrastructure management
SDDC Manager 9.0	192.168.1.241	Lifecycle management
Fleet Manager	192.168.1.78	VCF Operations lifecycle (fleet.lab.local)
NSX Manager VIP	192.168.1.70	Network virtualization
DNS/NTP Server	192.168.1.230	Name resolution and time sync

2.4 Appliance Services

Service	Description	Port	SystemD Unit
vPostgres Replication DB	PostgreSQL database	5433	vmware-vcops.service
GemFire Locator	Distributed cache locator	6061	vmware-vcops.service
Analytics	Main engine + GemFire cache server	10000	vmware-vcops.service
Collector	Data collection	—	vmware-vcops.service
API	REST API	—	vmware-vcops.service
CASA	Cluster and Slice Administration	—	vmware-casa.service
Web UI	Tomcat web interface	—	vmware-vcops-web.service
Watchdog	Service health monitor	—	vmware-vcops-watchdog.service
Apache HTTPD	HTTPS reverse proxy	443	httpd.service

3. Root Cause

A Windows Update on the management workstation (Dell Precision 7920) abruptly shut down the machine, which brought down the entire nested VCF 9.0.1 environment. When the workstation came back, the VCF Operations VM was damaged and needed to be redeployed.

The OVA was deployed three times during recovery:

1. **First deploy:** OVA was deployed via `ovftool` but the OVF network properties were not applied (`ovftool` reported the property keys did not exist). The VM booted with DHCP and got IP `192.168.1.67` instead of `192.168.1.77`.
2. **Second deploy:** Attempts to manually set the static IP via `systemd-networkd` configuration files were overwritten by the `vrops-init.service` on every reboot, which resets network config from OVF properties.
3. **Third deploy:** A fresh deployment was done. OVF properties were set using the `/opt/vmware/bin/ovfenv` tool from inside the VM after first boot. The static IP `192.168.1.77` finally stuck.

However, by this point the appliance had already completed its first-boot initialization using the DHCP IP `192.168.1.67`. All internal services — SSL certificates, GemFire, PostgreSQL, CASA, Apache — were built around the wrong IP.

4. Session 1 — Deployment and Network Configuration (April 8)

4.1 First Deployment Attempt — DHCP and Wrong IP

The OVA was deployed using `ovftool` from VMware Workstation:

```
# Run from any shell (Linux/macOS/Git Bash) – calls the ovftool binary on PATH
ovftool --noSSLVerify \
  --acceptAllEulas \
  --name="vcf-ops" \
  --diskMode=thin \
  --datastore="vcenter-cl01-ds-vsan01" \
  --network="vcenter-cl01-vds01-pg-vm-mgmt" \
  --prop:root_password="<your-password>" \
  --prop:timezone="America/New_York" \
  --prop:ipv4_type="Static" \
  --prop:ipv4_address="192.168.1.77" \
  --prop:ipv4_netmask="255.255.255.0" \
  --prop:ipv4_gateway="192.168.1.1" \
  --prop:DNS="192.168.1.230" \
  --prop:ipv6_type="Disabled" \
  "E:\VCF_OVAs\VMware-Cloud-Foundation-Operations-9.0.2.0-25073845_OVF10.ova" \
  "vi://administrator@vsphere.local@vcenter.lab.local/vcenter-dc01/host/vcenter-cl01/esxi04.lab.local"
```

PowerShell equivalent (run on your Windows workstation):

```
# PowerShell on Windows workstation – invokes ovftool.exe directly with backtick continuations
& 'C:\Program Files\VMware\VMware OVF Tool\ovftool.exe' `
  --noSSLVerify `
  --acceptAllEulas `
  --name=vcf-ops `
```

```
--diskMode=thin `
--datastore=vcenter-cl01-ds-vsan01 `
--network=vcenter-cl01-vds01-pg-vm-mgmt `
--prop:root_password='<your-password>' `
--prop:timezone=America/New_York `
--prop:ipv4_type=Static `
--prop:ipv4_address=192.168.1.77 `
--prop:ipv4_netmask=255.255.255.0 `
--prop:ipv4_gateway=192.168.1.1 `
--prop:DNS=192.168.1.230 `
--prop:ipv6_type=Disabled `
'E:\VCF OVA\VMware-Cloud-Foundation-Operations-9.0.2.0-25073845_OVF10.ova' `
'vi://administrator@vsphere.local@vcenter.lab.local/vcenter-dc01/host/vcenter-cl01/esxi04.lab.local'
```

Problem: ovftool reported that the network property keys did not exist. The `root_password` and `timezone` properties were accepted, but the network properties (`ipv4_type` , `ipv4_address` , etc.) were silently ignored. The VM booted with DHCP and picked up `192.168.1.67` .

4.2 Network Configuration Struggles

Multiple attempts were made to set the static IP by editing `/etc/systemd/network/10-eth0.network` directly:

```
[Match]
Name=eth0

[Network]
DHCP=no
Address=192.168.1.77/24
Gateway=192.168.1.1
DNS=192.168.1.230

[DHCP]
ClientIdentifier=mac
UseDomains=true
```

After each edit, `systemctl restart systemd-networkd` was run. The IP changed to `.77` temporarily, but reverted to DHCP after every reboot.

4.3 Discovery: vrops-init Overwrites Network Config on Boot

Investigation found that `vrops-init.service` runs at every boot and overwrites the network configuration based on OVF properties:

```
# Run on the appliance only (SSH as root) – systemctl is Photon OS init system
systemctl cat vrops-init.service
```

The service runs `/opt/vmware/bin/vrops_network_manager` which reads the OVF environment properties and regenerates the `systemd-networkd` configuration files. It also creates an override directory at `/etc/systemd/network/10-eth0.network.d/` that overrides any manual changes.

4.4 The ovfenv Tool — Setting Properties Correctly

The correct way to set the IP is through the OVF property store, not by editing network files directly. The properties are stored internally and read by `vrops-init.service` on boot.

The `ovfenv` tool at `/opt/vmware/bin/ovfenv` reads and writes these properties:

```
# Run on the appliance only (SSH as root) – /opt/vmware/bin/ovfenv ships with the OVA
# View all OVF properties
/opt/vmware/bin/ovfenv --dump
```

Before fix, the output showed:

```
[ipv4_type.VMware_Aria_Operations]=DHCP
[ipv4_address.VMware_Aria_Operations]=
[ipv4_netmask.VMware_Aria_Operations]=
[ipv4_gateway.VMware_Aria_Operations]=
[DNS.VMware_Aria_Operations]=
[ipv6_type.VMware_Aria_Operations]=
```

4.5 Second Deployment — Delete and Redeploy

The first VM was deleted from vCenter and a fresh deployment was attempted. This also failed because the vCenter inventory path was wrong:

```
Error: locator does not refer to an object
```

4.6 Third Deployment — ovftool With Correct Path

The correct vCenter path was determined by querying the API:

- Datacenter: `vcenter-dc01`
- Cluster: `vcenter-cl01`
- Host: `esxi04.lab.local`

The corrected deploy command:

```
# Run from any shell (Linux/macOS/Git Bash) – corrected vCenter inventory path
ovftool --noSSLVerify \
  --acceptAllEulas \
  --name="vcf-ops" \
  --diskMode=thin \
  --datastore="vcenter-cl01-ds-vsan01" \
  --network="vcenter-cl01-vds01-pg-vm-mgmt" \
  --prop:root_password="<your-password>" \
  --prop:timezone="America/New_York" \
  "E:\VCF_OVAs\VMware-Cloud-Foundation-Operations-9.0.2.0-25073845_OVF10.ova" \
  "vi://administrator@vsphere.local@vcenter.lab.local/vcenter-dc01/host/vcenter-cl01/esxi04.lab.local"
```

PowerShell equivalent (run on your Windows workstation):

```
# PowerShell on Windows workstation – same minus the network props (deliberately)
& 'C:\Program Files\VMware\VMware OVF Tool\ovftool.exe' `
  --noSSLVerify `
  --acceptAllEulas `
  --name=vcf-ops `
  --diskMode=thin `
  --datastore=vcenter-cl01-ds-vsan01 `
  --network=vcenter-cl01-vds01-pg-vm-mgmt `
  --prop:root_password='<your-password>' `
  --prop:timezone=America/New_York `
```

```
'E:\VCF OVAs\VMware-Cloud-Foundation-Operations-9.0.2.0-25073845_OVF10.ova' `
'vi://administrator@vsphere.local@vcenter.lab.local/vcenter-dc01/host/vcenter-cl01/esxi04.lab.local'
```

The VM deployed successfully but again booted with DHCP (IP 192.168.1.67) because the network properties were not included.

4.7 Setting OVF Properties and Static IP

After SSH was enabled from the console, the OVF properties were set from inside the VM:

```
# Run on the appliance only (SSH as root) - /opt/vmware/bin/ovfenv ships with the OVA
/opt/vmware/bin/ovfenv --key ipv4_type.VMware_Aria_Operations --value Static
/opt/vmware/bin/ovfenv --key ipv4_address.VMware_Aria_Operations --value 192.168.1.77
/opt/vmware/bin/ovfenv --key ipv4_netmask.VMware_Aria_Operations --value 255.255.255.0
/opt/vmware/bin/ovfenv --key ipv4_gateway.VMware_Aria_Operations --value 192.168.1.1
/opt/vmware/bin/ovfenv --key DNS.VMware_Aria_Operations --value 192.168.1.230
/opt/vmware/bin/ovfenv --key ipv6_type.VMware_Aria_Operations --value Disabled
```

Important: The valid value for IPv4 type is `Static` (capital S), not `STATIC` . This was confirmed by inspecting the OVF descriptor's ValueMap: `DHCP` , `Static` , `Disabled` .

The VM was then rebooted.

4.8 IP Confirmed at 192.168.1.77

After reboot, `vrops-init.service` read the OVF properties and configured the static IP correctly:

```
# Run on the appliance only (SSH as root) - iproute2 / Linux network stack
ip addr show eth0 | grep inet
# inet 192.168.1.77/24 brd 192.168.1.255 scope global eth0
```

The resulting network configuration:

```
# /etc/systemd/network/10-eth0.network
[Match]
Name=eth0

[Network]
DHCP=no
DNS=192.168.1.230

# /etc/systemd/network/10-eth0.network.d/ipv4.conf
[Network]
Address=192.168.1.77/24
Gateway=192.168.1.1
```

Result: Static IP 192.168.1.77 confirmed and persistent across reboots. However, all internal services had already been initialized with the DHCP IP 192.168.1.67 during first boot.

5. Session 2 — Hostname and DNS Fix (April 9)

5.1 Hostname Change

The default hostname `VMwareCloudFoundationOperationsClusterNode` needed to change to `vcf-ops.lab.local` :

```
# Run on the appliance only (SSH as root) – hostnamectl is systemd hostname tool
hostnamectl set-hostname vcf-ops.lab.local
```

Verification:

```
hostname -f
# vcf-ops.lab.local
```

5.2 Hosts File Update

```
# Run on the appliance only (SSH as root) – edits /etc/hosts in-place
sed -i 's/VMwareCloudFoundationOperationsClusterNode/vcf-ops.lab.local vcf-ops/g' /etc/hosts
```

Resulting `/etc/hosts` :

```
127.0.0.1 localhost
192.168.1.77 vcf-ops.lab.local vcf-ops
::1 vcf-ops.lab.local vcf-ops localhost ipv6-localhost ipv6-loopback
```

5.3 VMware Hostname Script

The appliance includes a built-in hostname script:

```
# Run on the appliance only (SSH as root) – script lives in /opt/vmware/bin/
python3 /opt/vmware/bin/set_hostname.py vcf-ops.lab.local
```

Result: Hostname changed successfully to `vcf-ops.lab.local` .

6. Session 2 — SSL Certificate Regeneration

6.1 Certificate Inventory

All SSL files are in `/storage/vcops/user/conf/ssl/` . A backup was created first:

```
# Run on the appliance only (SSH as root) – file edits in /storage/vcops/
cp -r /storage/vcops/user/conf/ssl /storage/vcops/user/conf/ssl-backup
```

File	Purpose	Action
cacert.pem	Self-signed Certificate Authority	No change needed
cakey.pem	CA private key	No change needed
slice_1_cert.pem	Main node certificate	Regenerated
slice_1_key.pem	Main node private key	Regenerated
slice_1_cert.pfx	PKCS12 bundle	Regenerated
tcserver.keystore	Java KeyStore for Tomcat	Rebuilt
tcserver.truststore	Java TrustStore for inter-service trust	Rebuilt
cluster.truststore	Java TrustStore for cluster communication	Rebuilt
jre_and_tcserver.truststore	Combined JRE + Tomcat TrustStore	Rebuilt
web_cert.pem	Symlink to slice_1_cert.pem	Auto-updated
web_key.pem	Symlink to slice_1_key.pem	Auto-updated
postgres_vcops_cert.pem	PostgreSQL client cert (CN=vcops)	Restored from backup
postgres_vcops_key.pem	PostgreSQL client key	Restored from backup

6.2 Keystore Password Discovery

Passwords were found in `/storage/vcops/user/conf/ssl/storePass.properties` :

```
sslkeystorePassword=KaQYLZxZ48HdiowMG0uUF0JTeJKu4n
ssltruststorePassword=nT3DlApTGJX/Mcm9n7DlH2K2wT5ftd
```

The keystore alias (`tcserver`) and type (`JKS`) were found in the Tomcat config at `/usr/lib/vmware-vcops/tomcat-enterprise/conf/catalina.properties` .

6.3 Comprehensive Certificate Regeneration Script

A script (`fix_certs.sh`) was uploaded via `pscp` and executed. It performed these steps:

Step 1 — Generate SAN config and new certificate:

```
# Run on the appliance only (SSH as root) – uses /storage/vcops/ CA key + /tmp scratch
printf '[v3_ext]\nsubjectAltName = DNS:vcf-ops.lab.local,DNS:localhost,IP:127.0.0.1,IP:192.168.1.77\n' >
/tmp/san.cnf

openssl req -new -newkey rsa:2048 -nodes \
  -keyout /tmp/new_key.pem -out /tmp/new.csr \
  -subj "/CN=vcf-ops.lab.local/O=Broadcom, Inc./OU=Broadcom, Inc."

openssl x509 -req -in /tmp/new.csr \
  -CA /storage/vcops/user/conf/ssl/cacert.pem \
  -CAkey /storage/vcops/user/conf/ssl/cakey.pem \
  -CAcreateserial -out /tmp/new_cert.pem -days 1825 \
  -extfile /tmp/san.cnf -extensions v3_ext
```

Step 2 — Replace PEM files with correct ownership and permissions.

Step 3 — Create PKCS12 bundle:

```
# Run on the appliance only (SSH as root) – uses /tmp scratch + ssl dir CWD
openssl pkcs12 -export -in /tmp/new_cert.pem -inkey /tmp/new_key.pem \
-certfile cacert.pem -out /tmp/new.p12 -name tcserver \
-passout "pass:KaqYLZxZ48HdiowMG0uUF0JTeJKu4n"
```

Step 4 — Rebuild tcserver.keystore:

```
# Run on the appliance only (SSH as root) – keytool is the bundled JRE; CWD is
/storage/vcops/user/conf/ssl/
rm -f tcserver.keystore
keytool -importkeystore -srckeystore /tmp/new.p12 -srcstoretype PKCS12 \
-srcstorepass "KaqYLZxZ48HdiowMG0uUF0JTeJKu4n" \
-destkeystore tcserver.keystore -deststoretype JKS \
-deststorepass "KaqYLZxZ48HdiowMG0uUF0JTeJKu4n" \
-destkeypass "KaqYLZxZ48HdiowMG0uUF0JTeJKu4n" \
-alias tcserver -noprompt
```

Steps 5-7 — Rebuild all three TrustStores (same command for each):

```
# Run on the appliance only (SSH as root) – keytool, CWD is /storage/vcops/user/conf/ssl/
rm -f tcserver.truststore
keytool -importcert -alias vcops-cluster-ca -file cacert.pem \
-keystore tcserver.truststore \
-storepass "nT3DlApTGJX/Mcm9n7DlH2K2wT5ftd" \
-storetype JKS -noprompt
```

Repeated for `cluster.truststore` and `jre_and_tcserver.truststore`.

Step 8 — Update PKCS12 file (`slice_1_cert.pfx`).

Step 9 — Restore PostgreSQL client certificates from backup (CN= `vcops` , did not need changing).

6.4 Apache HTTPD Certificate

Even after replacing all certificates in the SSL directory, the web server on port 443 still served the old certificate. Apache uses separate files:

- Config: `/etc/httpd/conf/extra/httpd-ssl.conf`
- Certificate: `/etc/httpd/conf/server.crt`
- Key: `/etc/httpd/conf/server.key`

```
# Run on the appliance only (SSH as root) – file edits in /etc/httpd/ + systemctl restart
cp -f /storage/vcops/user/conf/ssl/slice_1_cert.pem /etc/httpd/conf/server.crt
cp -f /storage/vcops/user/conf/ssl/slice_1_key.pem /etc/httpd/conf/server.key
systemctl restart httpd
```

6.5 vPostgres Server Certificate

The PostgreSQL server uses its own certificate at `/var/vmware/vpostgres/current/vpostgres_cert.pem` , configured in `postgresql.conf` . This was still serving the old IP.

A script (`fix_vpg_cert.sh`) generated a new certificate with the correct SAN and copied it into place with `postgres:users` ownership.

6.6 Full Certificate Scan

After all replacements, a full filesystem scan confirmed only the backup directory contained old certificates:

```
# Run on the appliance only (SSH as root) – recursive find on the Photon OS rootfs
find / -name "*.pem" -o -name "*.crt" 2>/dev/null | while read f; do
  openssl x509 -in "$f" -noout -ext subjectAltName 2>/dev/null | \
    grep -q "192.168.1.67" && echo "OLD IP FOUND IN: $f"
done
```

Output: OLD IP FOUND IN: /storage/vcops/user/conf/ssl-backup/slice_1_cert.pem

Result: All 11 active certificate files now contain `vcf-ops.lab.local` and `192.168.1.77` in the SAN field. Apache and vPostgres both serve correct certificates.

7. Session 2 — GemFire and PostgreSQL Configuration

7.1 GemFire Configuration Files

Four configuration files contained the old IP:

File	Original Setting
/usr/lib/vmware-vcops/user/conf/gemfire.properties	locators=192.168.1.67:6061
/usr/lib/vmware-vcops/user/conf/gemfire.native.properties	bind-address=192.168.1.67
/usr/lib/vmware-vcops/user/conf/gemfire.locator.properties	bind-address=192.168.1.67
/usr/lib/vmware-vcops/user/conf/persistence/persistence.properties	JDBC URLs with 192.168.1.67:5433

```
# Run on the appliance only (SSH as root) – file edits in /usr/lib/vmware-vcops/
sed -i 's/192.168.1.67/192.168.1.77/g' \
  /usr/lib/vmware-vcops/user/conf/gemfire.properties \
  /usr/lib/vmware-vcops/user/conf/gemfire.native.properties \
  /usr/lib/vmware-vcops/user/conf/gemfire.locator.properties \
  /usr/lib/vmware-vcops/user/conf/persistence/persistence.properties
```

7.2 PostgreSQL Database Updates

A script searched all text, JSON, JSONB, and bytea columns across all schemas in `vcopsdb`. Two locations contained the old IP:

Table	Column	Type	Rows
metric_key	metric_key	text	1
resource	resource_key	jsonb	10

```
-- Run as postgres user
UPDATE metric_key SET metric_key = replace(metric_key, '192.168.1.67', '192.168.1.77')
  WHERE metric_key LIKE '%192.168.1.67%';

UPDATE resource SET resource_key =
  replace(resource_key::text, '192.168.1.67', '192.168.1.77')::jsonb
  WHERE resource_key::text LIKE '%192.168.1.67%';
```

Result: All GemFire configs and PostgreSQL data now reference 192.168.1.77 .

8. Session 2 — CASA Cluster Membership

8.1 HSQLDB Update

CASA stores cluster membership in an HSQLDB database at `/storage/db/casa/webapp/hsqldb/casa.db.script` . The `clusterMembership` document was updated from `"ip_address": "192.168.1.67"` to `"ip_address": "192.168.1.77"` using a Python script that safely parsed the JSON inside the SQL INSERT statement.

8.2 The Unresolvable Admin Slice Problem

Despite updating every file, database, certificate, and configuration on the system, CASA's `MembershipHelper.findAdminSlice` method continued to resolve the admin slice to 192.168.1.67 .

Investigation performed: - Confirmed HSQLDB had 192.168.1.77 — verified after every CASA restart - Searched all PostgreSQL tables across all schemas, all column types including bytea — zero results - Wiped and rebuilt the entire GemFire data directory (`/storage/db/vcops/data`) — still .67 - Scanned every certificate on disk — none contained .67 (except backups) - Searched all configuration files in `/storage/` , `/opt/vmware/` , `/usr/lib/vmware-casa/` , `/etc/` — zero results - Searched all .dat files in GemFire persistent storage — zero results - Fully rebooted the VM — CASA still resolved .67 after a clean boot - Deleted the `renewalJWT.js` on file which contained the old certificate — still .67

Conclusion: The 192.168.1.67 IP is derived at runtime by CASA's Java code from a source that is not stored on disk in any human-readable or grep-searchable format. It is likely embedded in a serialized Java object or derived from an internal API call that caches the original deployment IP permanently.

9. Session 2 — IP Alias Workaround

Since CASA could not be made to use 192.168.1.77 as the admin slice IP through any configuration change, the solution was to make the appliance respond on 192.168.1.67 as well.

9.1 Add Secondary IP

```
# Run on the appliance only (SSH as root) – iproute2 / Linux network stack
ip addr add 192.168.1.67/24 dev eth0
```

Verification:

```
ip addr show eth0
# inet 192.168.1.77/24 brd 192.168.1.255 scope global eth0
# inet 192.168.1.67/24 scope global secondary eth0
```

9.2 Make It Persistent

Created a systemd-networkd drop-in configuration:

```
# Run on the appliance only – write this file content via vi / cat heredoc as root
# /etc/systemd/network/10-eth0.network.d/secondary-ip.conf
[Address]
Address=192.168.1.67/24
```

This ensures the secondary IP survives reboots.

9.3 Create DNS Record

A DNS A record was created for `192.168.1.67` so that Fleet Manager and other external systems can reach the VCF Operations node on the alias IP.

Result after adding the IP alias and rebooting: - All VCF Operations services started successfully - Analytics (GemFire cache server) started and port 10000 came up - CASA could reach itself on `192.168.1.67` and resolve the admin slice - The web UI became fully functional

10. Session 2 — Fleet Management Connection

10.1 Registration Flow and Failure

When clicking "Finish" in the Fleet Management wizard at `https://vcf-ops.lab.local/admin`, CASA:

1. Resolved the admin slice IP to `192.168.1.67` (via the IP alias)
2. Submitted a POST to `https://fleet.lab.local/lcm/service-account/api/service-registries/registrations`
3. Fleet Manager received the registration with `192.168.1.67` as the callback address
4. Fleet Manager tried to call back to `https://192.168.1.67/casa/deployment/slice/role/status`

Without the DNS record, step 4 failed with HTTP 500 because Fleet Manager could not resolve `192.168.1.67`. With the DNS record in place, Fleet Manager could reach the node.

10.2 SDDC Manager and NSX Issues

During the Fleet Management troubleshooting, two additional issues were encountered:

- **SDDC Manager** (`192.168.1.241`) experienced CPU issues and needed a reboot
- **NSX Manager** (`192.168.1.71`) experienced a CPU soft lockup and needed a reboot

Both were rebooted and recovered before the Fleet Management connection was retried.

10.3 Successful Connection

After SDDC Manager and NSX recovered, the Fleet Management wizard was retried:

1. Navigated to `https://vcf-ops.lab.local/admin`
2. Clicked "Connect" in the Fleet Management section
3. Entered the admin password (`<your-password>`)

4. Clicked "Finish"

Result: Fleet Management connected successfully. The admin page shows Fleet Management as connected. All adapters show "Data receiving" status.

11. Session 3 — VCF Cloud Account Configuration (April 9)

After Fleet Management was connected and all VCF Operations services were running, the next step was to add a VCF Cloud Account so VCF Operations could discover and monitor the entire VCF environment.

11.1 Add VCF Cloud Account

1. Navigated to **Administration > Integrations**
2. Clicked **ADD ACCOUNT** and selected **VMware Cloud Foundation**
3. Filled in the account details:

Field	Value
Name	VCf LAB
Description	VCf 9.0.1 Management Domain - lab.local
Physical Data Center	Lab-DC (created during this step)
SDDC Manager	sddc-manager.lab.local
Credential	administrator@vsphere.local
Collector / Group	Default collector group

4. Clicked **VALIDATE CONNECTION** — passed
5. Clicked **ADD**

VCf Operations automatically discovered the management domain and created adapters for vCenter, vSAN, and NSX.

Result: VCF Cloud Account `VCf LAB` created and collecting. vCenter adapter (`vcenter.lab.local`) and vSAN adapter both show **Collecting** status.

11.2 NSX Adapter Failure

The NSX adapter was created automatically but showed **Warning** status. Investigation of the collector logs revealed:

```
Failed to create service account: SvcAccountInfo(  
  client=ops, server=nsx, serverFqdn=nsx-vip.lab.local,  
  accountName=svc-ops-nsx-be018b3df3c34e9d-1,  
  role=enterprise_admin)
```

```
Exception occurred while creating service account. Error code: 500
API returned with unexpected status code 500
```

VCF Operations was trying to create a service account on NSX Manager VIP (`nsx-vip.lab.local`) but NSX was returning HTTP 500. The NSX VIP (`192.168.1.70`) was unreachable — ping from VCF Ops returned "Destination Host Unreachable."

The NSX node IP (`192.168.1.71`) was pingable, indicating the OS was running but NSX platform services were down. This was the same cold start issue documented in the NSX Manager Cold Start RCA.

12. Session 3 — NSX Manager Cold Start Recovery

12.1 Diagnosis

NSX Manager had experienced a CPU soft lockup earlier in the day and was rebooted. After the reboot, the platform services did not auto-start — the same cold start failure pattern documented previously.

SSH into the NSX Manager as `admin` and running `get cluster status` showed:

Group	Status
DATASTORE	STABLE / UP
CONTROLLER	STABLE / UP
MANAGER	UNAVAILABLE / DOWN
HTTPS	UNAVAILABLE / DOWN
IDPS_REPORTING	UNAVAILABLE / DOWN

The Corfu datastore and Controller were already running, but HTTP and Manager were down.

12.2 Service Recovery

Following the documented cold start recovery procedure, services were started in dependency order from the NSX CLI (`admin` user):

```
start service http
```

Wait 30 seconds

```
start service manager
```

Wait 60 seconds

After starting HTTP and Manager, all remaining services cascaded automatically. Verification:

```
get cluster status verbose
```

Result: Overall Status changed to **STABLE**. All groups showed **UP**:

Group	Status
DATASTORE	STABLE / UP
CONTROLLER	STABLE / UP
MANAGER	STABLE / UP
HTTPS	STABLE / UP
IDPS_REPORTING	STABLE / UP
CM-INVENTORY	STABLE / UP
MESSAGING-MANAGER	STABLE / UP
SITE_MANAGER	STABLE / UP
MONITORING	STABLE / UP
CORFU_NONCONFIG	STABLE / UP

12.3 NSX Adapter Success

After NSX Manager recovered, the NSX adapter in VCF Operations was manually added:

1. **Administration > Integrations > ADD ACCOUNT > NSX**
2. Filled in:

Field	Value
Name	NSX Manager
Description	NSX Manager 9.0.1 - lab.local
Virtual IP/NSX Manager	nsx-vip.lab.local
Credential Kind	NSX Credentials
Credential Name	NSX Admin
Username	admin
Collector / Group	Default collector group

3. Clicked **VALIDATE CONNECTION** — passed
4. Clicked **ADD**

The NSX adapter initially showed **Warning** while initializing, then transitioned to **Collecting** after approximately 10 minutes.

After all adapters were collecting, **Activate Management** was clicked at the top of the Integrations page. This greyed out after activation, confirming management mode was enabled.

Result: NSX Manager fully recovered. NSX adapter collecting. Management activated for the VCF Cloud Account.

13. Session 3 — VCF Operations for Logs Integration

13.1 Logs Appliance Verification

The VCF Operations for Logs appliance (`logs.lab.local` / `192.168.1.242`) was already deployed but not integrated with VCF Operations.

Version check:

```
# Run from any shell with curl on PATH (workstation, appliance, container) – VCF Operations for Logs REST API
curl -sk https://192.168.1.242:9543/api/v2/version
```

PowerShell equivalent (run on your Windows workstation):

```
# PowerShell on Windows workstation
[Net.ServicePointManager]::ServerCertificateValidationCallback = {$true}
[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12

Invoke-RestMethod -Uri 'https://192.168.1.242:9543/api/v2/version'
```

```
{"releaseName":"Nightly","version":"9.0.1.0.24960345"}
```

Port scan from VCF Ops node:

Port	Status	Purpose
514	OPEN	Syslog UDP/TCP
1514	OPEN	Syslog TLS
6514	OPEN	Syslog TLS (alternate)
9543	OPEN	CFAPI ingestion and API
443	OPEN	Web UI

13.2 Control Panel Integration

1. Navigated to **Administration > Control Panel**
2. Clicked **Operations-Logs Appliance Integration** tile
3. The integration was already registered and showed:

```
Address: 192.168.1.242
Status:  ✓ Connected
```

13.3 Enabling Log Collection via API

The Log Collection page at **Infrastructure Operations > Configurations > Log Collection** showed the EDIT button greyed out. This was because the vCenter adapter's `LOG_COLLECTION` property was set to `NONE`.

Fix via VCF Operations REST API:

```

# Authenticate
$token = Invoke-RestMethod -Uri 'https://vcf-ops.lab.local/suite-api/api/auth/token/acquire' ...

# Get vCenter adapter
$adapter = Invoke-RestMethod -Uri '.../api/adapters/ac2c287c-f1dc-4b20-97fa-61b6108c8f74' ...

# Change LOG_COLLECTION from NONE to LOGCLUSTER
foreach ($ri in $adapter.resourceKey.resourceIdentifiers) {
    if ($ri.identifierType.name -eq 'LOG_COLLECTION') {
        $ri.value = 'LOGCLUSTER'
    }
}

# Remove collectorId (causes 422 error if both collectorId and collectorGroupId present)
$adapter.PSObject.Properties.Remove('collectorId')
# Remove other read-only fields...

# PUT updated adapter
Invoke-RestMethod -Uri '.../api/adapters' -Method PUT -Body $json ...

# Restart adapter
Invoke-RestMethod -Uri '.../api/adapters/.../monitoringstate/stop' -Method PUT ...
Invoke-RestMethod -Uri '.../api/adapters/.../monitoringstate/start' -Method PUT ...

```

Result: `LOG_COLLECTION` changed to `LOGCLUSTER`. EDIT button became active after the next collection cycle.

13.4 vCenter Syslog Forwarding

vCenter was configured to forward logs to the Logs appliance via TLS on port 1514:

```

# Run from any shell with curl on PATH – vCenter REST API session + logging forwarding config
TOKEN=$(curl -sk -X POST "https://vcenter.lab.local/api/session" \
-u 'administrator@vsphere.local:<password>' | tr -d '"')

curl -sk -X PUT "https://vcenter.lab.local/api/appliance/logging/forwarding" \
-H "vmware-api-session-id: $TOKEN" \
-H "Content-Type: application/json" \
-d '{"cfg_list":[{"hostname":"192.168.1.242","port":1514,"protocol":"TLS"}]}'

```

PowerShell equivalent (run on your Windows workstation):

```

# PowerShell on Windows workstation – cert bypass already set above; reuse session
$vc      = 'vcenter.lab.local'
$vcUser  = 'administrator@vsphere.local'
$vcPass  = '<password>'
$pair    = [System.Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes("${vcUser}:${vcPass}"))
$session = Invoke-RestMethod -Method Post -Uri "https://$vc/api/session" `
    -Headers @{ Authorization = "Basic $pair" }
$TOKEN = $session # vCenter returns the session id as a bare quoted string

$body = @{
    cfg_list = @( @{ hostname = '192.168.1.242'; port = 1514; protocol = 'TLS' } )
} | ConvertTo-Json -Depth 5

Invoke-RestMethod -Method Put -Uri "https://$vc/api/appliance/logging/forwarding" `
    -Headers @{ 'vmware-api-session-id' = $TOKEN } `
    -ContentType 'application/json' -Body $body

```

Connection test:

```

# Run from any shell with curl on PATH – triggers a synthetic syslog test message from vCenter
curl -sk -X POST "https://vcenter.lab.local/api/appliance/logging/forwarding?action=test" \
-H "vmware-api-session-id: $TOKEN" \

```

```
-H "Content-Type: application/json" \  
-d '{"send_test_message":"true"}'
```

PowerShell equivalent (run on your Windows workstation):

```
# PowerShell on Windows workstation – reuses $vc and $TOKEN from previous block  
$testBody = @{ send_test_message = 'true' } | ConvertTo-Json  
Invoke-RestMethod -Method Post `\  
  -Uri "https://$vc/api/appliance/logging/forwarding?action=test" `\  
  -Headers @{ 'vmware-api-session-id' = $TOKEN } `\  
  -ContentType 'application/json' -Body $testBody
```

```
{ "hostname": "192.168.1.242", "state": "UP" }
```

13.5 Verification — Logs Flowing

Querying the Logs appliance API confirmed events were being received:

```
# Run from any shell with curl on PATH – VCF Operations for Logs CFAPI events query  
curl -sk "https://192.168.1.242:9543/api/v2/events?limit=5&order-by-direction=DESC" \  
-H "Authorization: Bearer $LOGS_TOKEN"
```

PowerShell equivalent (run on your Windows workstation):

```
# PowerShell on Windows workstation – assumes $LOGS_TOKEN was acquired earlier  
Invoke-RestMethod -Method Get `\  
  -Uri 'https://192.168.1.242:9543/api/v2/events?limit=5&order-by-direction=DESC' `\  
  -Headers @{ Authorization = "Bearer $LOGS_TOKEN" } |  
  ConvertTo-Json -Depth 10
```

```
{ "complete": true, "duration": 138, "numResults": 5, "events": [ ... ] }
```

Five events from `vcenter.lab.local` confirmed in the Logs appliance. Events were also visible in the Logs web UI at `https://logs.lab.local`.

13.6 Known Issue — UI Save Error

The VCF Operations Log Collection UI (**Infrastructure Operations > Configurations > Log Collection**) returned **"Error Saving, Please try again later"** when attempting to save the log collection configuration with TCP/SSL. Saving with UDP succeeded but then showed "Log Collection Stopped" because vCenter syslog was configured for TLS/1514.

This is a known VCF 9.0.1/9.0.2 UI bug. The underlying log pipeline works correctly — vCenter logs flow to the Logs appliance via direct syslog forwarding regardless of the UI configuration state.

Result: VCF Operations for Logs integration complete. vCenter logs flowing via TLS/1514. Logs searchable at `https://logs.lab.local`. UI save bug does not affect log delivery.

14. Complete List of Files Modified

Configuration Files

File Path	Change
/etc/hostname	Changed to <code>vcf-ops.lab.local</code>
/etc/hosts	Updated with <code>192.168.1.77 vcf-ops.lab.local vcf-ops</code>
/usr/lib/vmware-vcops/user/conf/gemfire.properties	Locator IP to <code>.77</code>
/usr/lib/vmware-vcops/user/conf/gemfire.native.properties	Bind address to <code>.77</code>
/usr/lib/vmware-vcops/user/conf/gemfire.locator.properties	Bind address to <code>.77</code>
/usr/lib/vmware-vcops/user/conf/persistence/persistence.properties	JDBC URLs to <code>.77</code>
/storage/db/casa/webapp/hsqldb/casa.db.script	Cluster membership IP to <code>.77</code>
/etc/systemd/network/10-eth0.network.d/secondary-ip.conf	New file — secondary IP alias <code>.67</code>

Certificate Files

File Path	Change
/storage/vcops/user/conf/ssl/slice_1_cert.pem	Regenerated with <code>.77</code> in SAN
/storage/vcops/user/conf/ssl/slice_1_key.pem	New private key
/storage/vcops/user/conf/ssl/slice_1_cert.pfx	Regenerated PKCS12 bundle
/storage/vcops/user/conf/ssl/tcserver.keystore	Rebuilt JKS with new cert
/storage/vcops/user/conf/ssl/tcserver.truststore	Rebuilt with CA cert
/storage/vcops/user/conf/ssl/cluster.truststore	Rebuilt with CA cert
/storage/vcops/user/conf/ssl/jre_and_tcserver.truststore	Rebuilt with CA cert
/etc/httpd/conf/server.crt	Replaced with new slice cert
/etc/httpd/conf/server.key	Replaced with new slice key
/var/vmware/vpostgres/current/vpostgres_cert.pem	Regenerated with <code>.77</code> in SAN
/var/vmware/vpostgres/current/vpostgres_key.pem	New private key

Database Changes

Database	Table.Column	Change
vcopsdb	<code>metric_key.metric_key</code>	Replaced <code>.67</code> with <code>.77</code> (1 row)
vcopsdb	<code>resource.resource_key</code>	Replaced <code>.67</code> with <code>.77</code> (10 rows)
HSQldb	<code>CASA_DOCS.clusterMembership</code>	IP and preferred_addresses to <code>.77</code>

OVF Properties Set

Property	Value
ipv4_type.VMware_Aria_Operations	Static
ipv4_address.VMware_Aria_Operations	192.168.1.77
ipv4_netmask.VMware_Aria_Operations	255.255.255.0
ipv4_gateway.VMware_Aria_Operations	192.168.1.1
DNS.VMware_Aria_Operations	192.168.1.230
ipv6_type.VMware_Aria_Operations	Disabled

15. Post-Repair Verification Checklist

Check	Command	Expected Result	Status
Hostname	<code>hostname -f</code>	<code>vcf-ops.lab.local</code>	PASS
Static IP	<code>ip addr show eth0</code>	<code>192.168.1.77/24</code>	PASS
IP alias	<code>ip addr show eth0 \ grep .67</code>	Secondary IP <code>.67</code> present	PASS
All services	<code>/opt/vmware/bin/vrops-status</code>	All running including Analytics	PASS
GemFire cache	<code>ss -tlnp \ grep 10000</code>	vcops-analytics listening	PASS
SSL cert	<code>openssl s_client -connect localhost:443</code>	CN=vcf-ops.lab.local, SAN <code>.77</code>	PASS
Web UI	<code>https://vcf-ops.lab.local</code>	Login page loads	PASS
Admin page	<code>https://vcf-ops.lab.local/admin</code>	Cluster Online, Fleet Connected	PASS
Adapters	Admin page adapter list	All show "Data receiving"	PASS
VCF Cloud Account	Administration > Integrations	VCF LAB — Collecting	PASS
vCenter adapter	Administration > Integrations	vcenter.lab.local — Collecting	PASS
vSAN adapter	Administration > Integrations	vcenter.lab.local - vSAN — Collecting	PASS
NSX adapter	Administration > Integrations	NSX Manager — Collecting	PASS
Management activated	Activate Management button	Greyed out (activated)	PASS
NSX cluster	<code>get cluster status</code> on NSX	Overall: STABLE	PASS
Logs appliance	<code>curl -sk https://192.168.1.242:9543/api/v2/version</code>	Version response	PASS
Logs integration	Control Panel > Ops-Logs Integration	Connected to 192.168.1.242	PASS
vCenter syslog	vCenter API forwarding test	state: UP	PASS
Logs flowing	Logs API events query	numResults > 0	PASS

Check	Command	Expected Result	Status
Logs UI	<code>https://logs.lab.local</code>	vCenter events visible	PASS

16. Lessons Learned

16.1 ovftool Does Not Always Apply OVF Network Properties

When deploying the VCF Operations OVA with `ovftool`, the network-related OVF properties (`ipv4_type`, `ipv4_address`, etc.) were silently ignored. The only reliable way to set them was from inside the VM using the `/opt/vmware/bin/ovfenv` tool after first boot.

16.2 vrops-init.service Overwrites Network Config on Every Boot

Editing `/etc/systemd/network/10-eth0.network` directly does not work. The `vrops-init.service` regenerates the network configuration from OVF properties on every boot. The correct approach is to set the OVF properties using `/opt/vmware/bin/ovfenv`, then reboot.

16.3 First Boot Initializes Everything with the Current IP

The appliance generates certificates, database entries, Java KeyStores, GemFire configurations, and cluster membership all during first boot. If the IP is wrong at first boot (for example, DHCP instead of static), all internal services are built around the wrong IP.

16.4 The IP is Stored in At Least 10 Different Locations

Changing the IP post-deployment requires updating PEM files, JKS keystores, JKS truststores, GemFire properties, PostgreSQL config, PostgreSQL database rows, CASA HSQLDB, Apache HTTPD cert, and vPostgres server cert. Missing any one causes SSL handshake failures or service crashes.

16.5 CASA Has an Unmodifiable Admin Slice Resolution

Despite updating every file, database, and certificate on the system, CASA's `MembershipHelper.findAdminSlice` continued to resolve the admin slice to the original deployment IP. The workaround was to add the original IP as a secondary address on the network interface and create a DNS record for it.

16.6 Always Back Up Before Making Changes

The first action taken was backing up the SSL directory. This backup was used multiple times when changes caused services to crash (for example, when only the keystore was updated but not the truststores).

16.7 SSH Password with ! Characters Requires Special Handling

When using PuTTY's `plink.exe` via bash to SSH into VCF appliances, passwords containing `!` characters get mangled by bash's history expansion. The fix is to use PowerShell instead of bash, or to run `set +H` before the `plink` command.

16.8 NSX Cold Start Is a Recurring Issue in Nested Labs

After any power event (crash, reboot, CPU lockup), NSX Manager's platform services may not auto-start. The recovery procedure is always the same: start HTTP, then Manager, then Controller — in that order from the NSX CLI

as the `admin` user.

16.9 Log Collection EDIT Button Requires API Change

The VCF Operations Log Collection page EDIT button is greyed out until the vCenter adapter's `LOG_COLLECTION` property is changed from `NONE` to `LOGCLUSTER` via the REST API. This property is not exposed in the UI.

16.10 VCF 9 Log Collection UI Has a Known Save Bug

The Log Collection configuration page returns "Error Saving" when trying to save with TCP/SSL. Logs flow correctly through direct syslog forwarding regardless of the UI state.

17. Document Information

Field	Value
Document Title	VCF Operations Node — Full Recovery from Windows Update Crash
Version	2.1
Author	Virtual Control LLC
Date	April 8–10, 2026 (rev May 7, 2026)
Classification	Internal — Lab Environment
Sessions	3 (evening of April 8, morning/afternoon of April 9, evening of April 9–10)
Total Duration	Approximately 10 hours
Deployments	3 attempts
Total Phases	13
Total Files Modified	19
Total Database Updates	3 tables
API Changes	vCenter adapter LOG_COLLECTION, vCenter syslog forwarding
Final Status	All services running, Fleet connected, Cloud Account collecting, NSX recovered, Logs integrated

Revision History

Version	Date	Notes
2.0	April 10, 2026	Initial release of full-recovery RCA covering deploy, network, certs, GemFire/PostgreSQL, CASA workaround, Fleet connect, Cloud Account, NSX cold-start recovery, Logs integration.
2.1	May 7, 2026	Option B PowerShell treatment: added top-of-doc callout linking to <code>../POWERSHELL-TRANSLATION-REFERENC</code> NCE.md ; added paste-ready PowerShell sibling blocks for workstation-runnable bash blocks (ovftool deploy commands, VCF Operations for Logs API version check, vCenter <code>/api/session</code> token + <code>/api/appliance/logging/forwarding</code> PUT/test, Logs <code>/api/v2/events</code> query); applied "Run on the appliance only" context comments to bash blocks targeting the Photon shell (<code>systemctl</code> , <code>/opt/vmware/bin/ovfenv</code> , <code>hostnamectl</code> , <code>sed -i /etc/hosts</code> and <code>/usr/lib/vmware-vcops/</code> , <code>openssl/keytool</code> cert ops in <code>/storage/vcops/user/conf/ssl/</code> , Apache HTTPD restart, recursive <code>find / , ip addr add , systemd-</code>

Version	Date	Notes
		networkd drop-in). PS 5.1 cert-bypass header inserted once at first PS sibling. SQL block left as psq1 per skip rules. No content/procedure changes.